

Qatar-1b

R-2816

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_210808 · created 2026-09-05T02:00:29+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459434.8419 +/- 0.0022 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.186 +/- 0.055
Transit depth (Rp/Rs)^2	3.44 +/- 2.04 [%]
Orbital Inclination (inc)	82.09 +/- 2.86
Airmass coefficient 1 (a1)	0.58 +/- 0.2
Airmass coefficient 2 (a2)	0.48 +/- 0.57
Scatter in the residuals of the lightcurve fit is	34.96 %
Best Comparison Star	#1 - [31, 94]
Optimal Aperture	3.78
Optimal Annulus	9.07
Transit Duration (day)	0.051 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.186 ± 0.055 vs 0.1463 (deviation 0.54σ)
Duration fit vs published	0.051 d vs 0.0692 d (deviation 0.49σ)
Mid-time O–C	5.3 min
Depth SNR	2.5
Residual scatter	34.96 %

Light curve

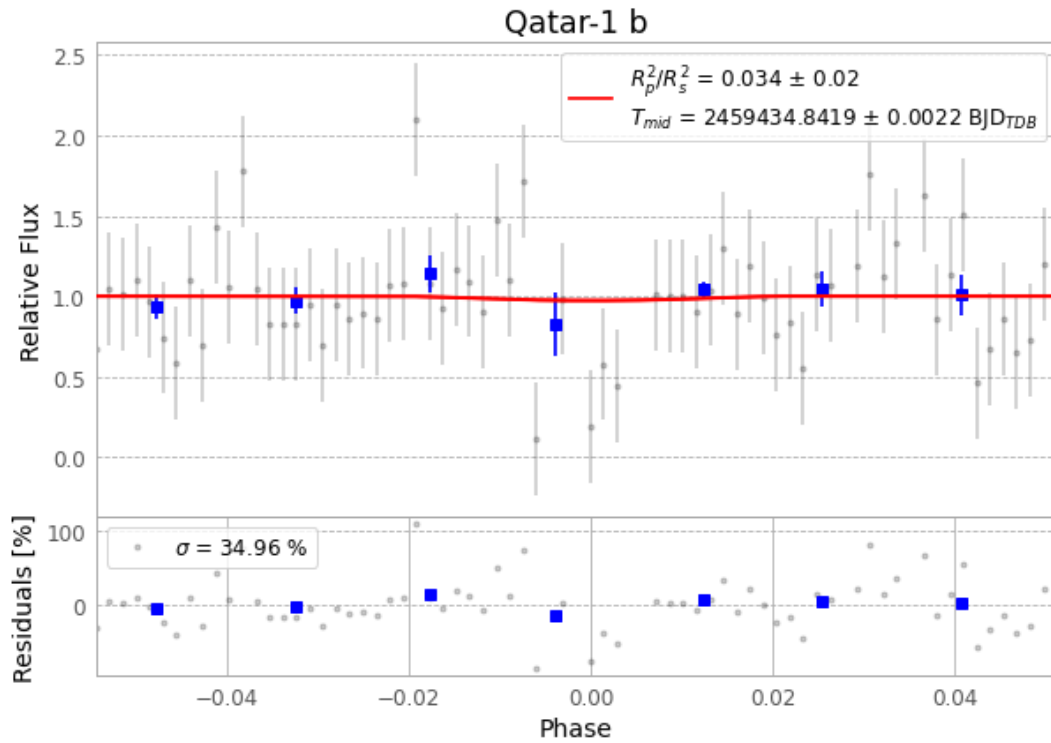


Figure 1 — FinalLightCurve_Qatar-1 b_2021-08-07.png (SHA-256 91f221255997d8f5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [453, 242], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 a2329dba3efe1f4c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

73 FITS frames (48 MB), Qatar-1210808061812.FITS → Qatar-1210808095412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-210808.log (29e941229145...), FinalLightCurve_Qatar-1 b_2021-08-07.png (91f221255997...), FinalLightCurve_Qatar-1 b_2021-08-07.csv (8e1b7bb83c58...)

Compute resources

Wall time 471.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-05 02:00Z.